Cheatsheet AZ-104 — 08 · Virtual machines: creation, sizes, disks, encryption, moving and availability

1. Creating a VM: what each tab decides

Tab	Decisions	Exam details
Basics	Image (Marketplace, **Azure Compute Gallery**, your own VHD), **size**, **zone** or availability set, user and authentication (SSH / password), inbound ports	Gen 1 vs **Gen 2** (Gen 2: Trusted Launch, OS disks > 2 TiB, UEFI). **Trusted launch** (Secure Boot + vTPM) is the default for Gen 2
Disks	OS disk type, encryption, data disks, "Delete with VM"	Premium SSD requires a size with **"s"** (e.g. D4**s**_v5)
Networking	VNet, subnet, **public IP**, NSG (on the NIC), load balancer	An NSG on the NIC and another on the subnet are both evaluated. No public IP → Bastion
Management	Identity (managed identity), **Entra login**, auto-shutdown, **backup**, patch orchestration, hotpatch	"Sign in with Entra credentials" is enabled here (extension + Virtual Machine Administrator / User Login role)
Monitoring	**Boot diagnostics** (managed or your own account), guest OS metrics, alerts	Serial console needs boot diagnostics
Advanced	Extensions, **custom data / user data** (cloud-init), VM applications, dedicated host, proximity placement group, capacity reservation	cloud-init is Linux only and first boot only

- Creation generates **VM, OS disk, NIC, public IP, NSG** and, if none exists, a VNet. Deleting the VM **does not delete** the NIC, disks or PIP unless "Delete with VM" is checked. **VM names**: 15 characters on Windows, 64 on Linux; a VM cannot be **renamed** (recreate it).
- **Images:** Marketplace (some with a paid plan) · **Azure Compute Gallery** (image definitions with **versions**, generalized (sysprep / `waagent –deprovision`) or **specialized**, replicated to several regions and shareable with other tenants or the community) · managed image (one region, no versions) · **VM Image Builder** to build them. For "deploy 50 identical VMs in 3 regions from a custom image" → Compute Gallery.
- **Sizes:** family by letter: **B** burstable (credits; dev/test), **D** general purpose, **E** memory, **F** CPU, **L** local storage, **M** huge memory, **N** GPU, **H** HPC. Suffixes: **s** Premium Storage, **a** AMD, **p** ARM, **d** local temp disk, **l** low memory, **m** more memory, **i** isolated, **v5** generation. **vCPU quota per family and region** → "QuotaExceeded" is fixed by requesting an increase, not by changing subscription.
- **Resizing:** the VM **restarts**; if the new size is not listed it is not available in the current cluster → **Stop (deallocate)** and check again; in an **availability set** you may have to deallocate **all** the VMs in the set. Windows cannot go from a size with a temp disk to one without (Linux can). The **temp disk** contents are lost.

2. Disks

Type	Purpose	Max IOPS / MB/s	Max size	OS disk
Standard HDD	Backup, dev/test, low access	2,000 / 500	32 TiB	Yes (**to be retired** as OS disk in Sept. 2028)
Standard SSD	Web, light apps, consistency	6,000 / 750	32 TiB	Yes
Premium SSD	Production, low latency; fixed **P1-P80** sizes, bursting, performance tiers (bump P30 → P40 without resizing)	20,000 / 900	32 TiB	Yes (needs an "s" VM)
Premium SSD v2	Production with adjustable IOPS and throughput **without changing the size**, and cheaper	80,000 / 2,000	64 TiB	**No**; data only; **zonal**; no host caching; not in Compute Gallery
Ultra Disk	SAP HANA, top-tier databases; IOPS and MB/s configurable on the fly	400,000 / 10,000	64 TiB	**No**; data only; no caching; **not in availability sets**; LRS only

- Roles: **OS** (Windows 127 GiB by default, up to 4 TiB), **temporary** (D: / `/dev/sdb`, **lost** on deallocate, resize or redeploy: never data), **data** (maximum number depends on the VM size). **Host caching**: None / ReadOnly / ReadWrite (default OS ReadWrite, data ReadOnly; None for Ultra and v2).
- Operations: **attach** new or existing (a disk on only one VM except **shared disks** for clusters), **detach**, **expand** (never shrink; data disks online in many cases, OS with the VM deallocated), **change type** Standard ↔ Premium (VM **deallocated**; to v2 / Ultra only by creating a new one), **snapshot** (full or **incremental**, copied to another region; create a disk or VM from it), **swap OS disk**, **export** via SAS, **managed image**. A disk must be in the **same region and zone** as the VM.

Encryption	What it encrypts	Keys	Requirements
SSE with platform-managed keys (default, always)	OS and data at rest	Microsoft	None
SSE with customer-managed keys	OS and data	Your Key Vault via **Disk Encryption Set**	Key Vault with soft delete and purge protection; the DES needs access to the key
Encryption at host (recommended)	OS, data, **temp disk and caches**, and VM ↔ storage traffic	Platform or CMK (DES)	Register the `EncryptionAtHost` feature in the subscription, **VM deallocated** to enable it, compatible sizes, **incompatible with ADE** on the same VM
Azure Disk Encryption (ADE)	Inside the VM with **BitLocker** / **dm-crypt**; OS, data and temp	**Key Vault** (+ optional KEK), "Enabled for disk encryption"	Uses VM CPU; not on Basic/A-series or with < 2 GB RAM; no custom Linux images; **to be retired in Sept. 2028** → migrate to encryption at host
Confidential disk encryption	OS disk bound to the vTPM	DES	Confidential VMs only (DCasv5 / ECasv5)

- "Encrypt the temp disk and caches with minimal effort" → encryption at host. "Keys must be in our Key Vault" → CMK (DES) or ADE. "Defender says disk encryption unhealthy" → enable encryption at host or ADE.

3. Availability and SLA

Option	What it is	SLA	Rules
Single VM	No redundancy	99.9 % with Premium SSD / Ultra · 99.5 % Standard SSD · 95 % Standard HDD	The SLA depends on the **OS disk**
Availability set	Spreads VMs across **fault domains** (up to **3**: power and switch) and **update domains** (up to **20**, default 5: maintenance reboots) within a datacenter	**99.95 %** (≥ 2 VMs)	Chosen **when creating** the VM (cannot be added later); same region; aligned with managed disks; does not combine with zones; recommended when the region **has no zones**
Availability zones	**Zonal** VM (1, 2 or 3), separate datacenters with their own power and network	**99.99 %** (≥ 2 VMs in ≥ 2 zones)	**Standard** public IP and load balancer (zone-redundant); a VM is not "moved" to a zone: recreate from snapshot / Backup, zone-to-zone Site Recovery or Resource Mover
VMSS	Automatic scaling (cheatsheet 09)	99.95 % (FD) · 99.99 % (zones)	Flexible spreads across zones and FDs
Proximity placement group	VMs in the same datacenter for **minimum latency**	—	Reduces availability; a PPG can span only one availability zone
Dedicated host	A physical server just for you	99.99 % with ≥ 2 hosts	Compliance, per-host licensing, isolation
Capacity reservation	Guarantees capacity for a size in a region / zone	—	For DR or planned peaks
Spot VM	Spare capacity up to ~90 % cheaper, **evictable** with 30 s notice (deallocate or delete)	**No SLA**	Batch, test, rendering; never critical production

- "99.99 % for a two-tier app" → VMs of each tier in **2 or more zones** behind a Standard Load Balancer. "99.95 %, region without zones" → availability set. VMs in an availability set must be created with the set from the start: "add an existing VM to the set" = recreate it.

4. Moving a VM

Destination	How	Conditions
Another resource group	Move (portal / `Move-AzResource` / `az resource move`)	Move the VM with **NIC and disks**; the VNet can stay; resources not in Succeeded state block it; the resource's RBAC assignments are lost; no ReadOnly lock
Another subscription	Move with **all dependents in the same RG**: VM, disks, NIC, PIP, NSG and **VNet**	Same tenant; provider registered at the destination; quota; **stop backup** (Recovery Services) and ASR first
Another region	**Azure Resource Mover** (wizard: dependencies, prepare, initiate, commit) or **Site Recovery** (replicate and fail over)	There is no native cross-region "move"; copies are created and the source is deleted
A zone	Site Recovery (zone to zone), Resource Mover or snapshot → zonal disk → new VM	—
An availability set	Recreate the VM (same NIC and OS disk are fine)	An existing VM cannot be added
Another tenant	Transfer the **subscription** (RBAC is lost)	—

5. Day-to-day operations

Operation	What it does	Watch out
Stop (from the OS) vs **Stop (deallocate)** (portal / CLI)	OS shutdown: the VM keeps **billing** compute · Deallocate: releases the host, **no compute billing** (disks still billed)	Deallocate loses the temp disk and the **dynamic public IP** (use static if you need it fixed)
Restart	Reboot without changing host	—
Redeploy	Moves the VM to **another host** (RDP / host connectivity problems)	Loses the temp disk; the dynamic IP may change
Reapply	Reapplies the configuration when the state is "Failed"	—
Boot diagnostics	Screenshot and boot log	Managed (recommended) or your own storage account
Serial console	Console over the serial port, no network	Needs boot diagnostics; Windows requires SAC
Run command	Run PowerShell / shell **without RDP or network** (`az vm run-command invoke --command-id RunPowerShellScript --scripts "..."`)	Useful to repair firewall / RDP
Reset password / **Reset SSH key**	Help → Reset password (VMAccess extension)	Also "Reset configuration only" for RDP
Bastion	RDP / SSH through the browser without a public IP	Cheatsheets 09 and 13
Auto-shutdown	Shuts down at a set time (DevTest) with notification	Only shuts down; does not start (use Automation / Start-Stop v2 for that)
Azure Update Manager	OS patching; patch orchestration **AutomaticByPlatform**, hotpatch (Windows Server Azure Edition)	Replaces Automation's Update Management
Azure Hybrid Benefit	`--license-type Windows_Server` (or RHEL / SLES) to use your own licenses with Software Assurance	Up to 40 % on Windows; also SQL on VM
Hibernation	Saves RAM to the OS disk and deallocates	Specific sizes and OSes

6. Commands that show up

Task	PowerShell (Az.Compute)	Azure CLI
Quick VM create	`New-AzVM -ResourceGroupName RG1 -Name vm1 -Location westeurope -Image Win2022Datacenter -Size Standard_D2s_v5 -Credential (Get-Credential) -Zone 1 -OpenPorts 3389`	`az vm create -g RG1 -n vm1 --image Ubuntu2204 --size Standard_D2s_v5 --admin-username azureuser --generate-ssh-keys --zone 1 --nsg-rule SSH`
Create with detailed config	`New-AzVMConfig -VMName vm1 -VMSize ...` → `Set-AzVMOperatingSystem` → `Set-AzVMSourceImage` → `Add-AzVMNetworkInterface` → `Set-AzVMOSDisk` → `New-AzVM -VM $vm`	`az vm create ... --vnet-name vnet1 --subnet sub1 --public-ip-address "" --availability-set as1 --os-disk-size-gb 256 --data-disk-sizes-gb 128 --license-type Windows_Server`
Status and sizes	`Get-AzVM -Status` · `Get-AzVMSize -Location westeurope` · `Get-AzVMSize -ResourceGroupName RG1 -VMName vm1` (those in the cluster)	`az vm list -d -o table` · `az vm list-sizes -l westeurope` · `az vm list-vm-resize-options -g RG1 -n vm1`
Start, stop, redeploy	`Start-AzVM` · `Stop-AzVM -Force` (deallocates) · `Stop-AzVM -StayProvisioned` (shuts down only) · `Restart-AzVM` · `Set-AzVM -Redeploy`	`az vm start` · `az vm deallocate` · `az vm stop` (shuts down without releasing) · `az vm restart` · `az vm redeploy`
Resize	`$vm = Get-AzVM ...; $vm.HardwareProfile.VmSize = "Standard_D4s_v5"; Update-AzVM -VM $vm -ResourceGroupName RG1`	`az vm resize -g RG1 -n vm1 --size Standard_D4s_v5`
Disks	`Add-AzVMDataDisk -VM $vm -Name d1 -Lun 0 -CreateOption Empty -DiskSizeInGB 256 -StorageAccountType Premium_LRS; Update-AzVM` · `New-AzSnapshot` · `New-AzDiskUpdateConfig -DiskSizeGB 512` + `Update-AzDisk`	`az vm disk attach -g RG1 --vm-name vm1 --name d1 --new --size-gb 256 --sku Premium_LRS` · `az vm disk detach` · `az snapshot create --source d1` · `az disk update -n d1 --size-gb 512` · `az disk update --sku Premium_LRS`
Encryption	`Set-AzVMDiskEncryptionExtension -ResourceGroupName RG1 -VMName vm1 -DiskEncryptionKeyVaultUrl ... -DiskEncryptionKeyVaultId ...` (ADE) · `Update-AzVM ... -EncryptionAtHost $true`	`az vm encryption enable -g RG1 -n vm1 --disk-encryption-keyvault kv1` (ADE) · `az vm update -g RG1 -n vm1 --set securityProfile.encryptionAtHost=true`
Availability set	`New-AzAvailabilitySet -ResourceGroupName RG1 -Name as1 -Location westeurope -PlatformFaultDomainCount 2 -PlatformUpdateDomainCount 5 -Sku Aligned`	`az vm availability-set create -g RG1 -n as1 --platform-fault-domain-count 2 --platform-update-domain-count 5`
Image and gallery	`New-AzGallery` · `New-AzGalleryImageDefinition` · `New-AzGalleryImageVersion -TargetRegion ...` · generalize: `Set-AzVm -Generalized`	`az sig create` · `az sig image-definition create` · `az sig image-version create --target-regions westeurope northeurope` · `az vm generalize`
Move	`Move-AzResource -DestinationResourceGroupName RG2 -ResourceId <vm>,<nic>,<disk>`	`az resource move --destination-group RG2 --ids <vm> <nic> <disk>`
Run command and access	`Invoke-AzVMRunCommand -ResourceGroupName RG1 -VMName vm1 -CommandId RunPowerShellScript -ScriptPath fix.ps1` · `Set-AzVMAccessExtension` (reset password)	`az vm run-command invoke -g RG1 -n vm1 --command-id RunShellScript --scripts "sudo apt update"` · `az vm user update -g RG1 -n vm1 -u azureuser -p <pwd>`

7. Decision tree

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What does the wording ask for?
├─ 99.99 % with VMs ..................................................... Availability zones (≥ 2 zones) + Standard LB
├─ 99.95 %, or region without zones ..................................... Availability set (≥ 2 VMs)
├─ Minimum latency between VMs .......................................... Proximity placement group
├─ Dedicated physical server / compliance ............................... Dedicated host
├─ Interruptible workload at minimum cost ............................... Spot VM
├─ Very fast OS disk .................................................... Premium SSD (size with "s")
├─ Adjustable IOPS without resizing, data ............................... Premium SSD v2 (or Ultra if extreme)
├─ Encrypt temp disk and caches ......................................... Encryption at host
├─ BitLocker / dm-crypt with Key Vault .................................. Azure Disk Encryption (retired 2028)
├─ Your own keys for the disks without touching the VM .................. SSE with CMK (Disk Encryption Set)
├─ Size not listed when resizing ........................................ Deallocate and retry
├─ Move to another region ............................................... Azure Resource Mover / Site Recovery
├─ Move to another subscription ......................................... Move with VM + disks + NIC + VNet in the same RG
├─ Put an existing VM into an availability set ........................... Recreate the VM
├─ Host problem, RDP not responding ..................................... Redeploy (or Run command / Serial console)
└─ No compute charges while stopped ..................................... Stop (deallocate), with a static IP if needed
```

8. Signals → answer

Wording	Answer
"VM stopped from the OS still incurs compute charges"	Stop (deallocate) from Azure
"public IP changes after each restart"	**Static** public IP (Standard)
"data on the temporary disk was lost"	Normal: D: / sdb is lost on deallocate, resize or redeploy
"the new size is not listed"	Deallocate first (cluster)
"resize a VM in an availability set"	May require deallocating all the VMs in the set
"attach Premium SSD fails"	The size is not "s" (Premium Storage capable)
"Ultra disk / Premium SSD v2 as OS disk"	Impossible: data only
"encrypt temp disk", "end-to-end encryption"	Encryption at host (feature registered, VM deallocated)
"ADE and encryption at host on the same VM"	Incompatible
"Key Vault for disk encryption not enabled"	Enable "Enabled for disk encryption" (ADE)

Wording	Answer
"deploy the same custom image to several regions"	Azure Compute Gallery with version replication
"generalize the VM before capturing"	sysprep /generalize (Windows) · `waagent -deprovision+user` (Linux) → `az vm generalize`
"specialized image keeps the computer name and accounts"	Specialized image in the gallery
"move VM to another subscription" fails	Dependents not moved (VNet, disks) or backup / ASR active
"move VM to another region"	Resource Mover (or Site Recovery)
"add VM to availability zone after creation"	Recreate (snapshot) or zone-to-zone Site Recovery
"Windows VM cannot change to a size without temp disk"	Limitation: create a new VM from a snapshot
"use existing Windows Server licenses"	Azure Hybrid Benefit (`--license-type Windows_Server`)
"connect without public IP"	Bastion (or Run command / Serial console to repair)
"boot problem, see the screen"	Boot diagnostics screenshot + Serial console

9. Mistakes to avoid

1. Storing data on the temp disk.
2. Choosing a size without "s" and expecting Premium disks.
3. Assuming a VM can be added to an availability set or a zone after creation.
4. Confusing the OS Stop with Stop (deallocate): only the latter stops billing compute.
5. Combining ADE with encryption at host, or choosing ADE for new projects (retired in 2028).
6. Moving to another subscription without the dependent resources in the same RG, or with backup active.
7. Expecting a native cross-region "move": it is Resource Mover or Site Recovery.
8. Using Spot for production, or Standard HDD as OS disk in new designs.
9. Forgetting that resizing restarts the VM and that a Premium SSD v2 or Ultra disk is zonal and has no cache.